

Enhanced Anomaly Detection in Solar Power Plants Using Hybrid Machine Learning Techniques

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Abstract—Anomalies in solar power plants, such as equipment malfunctions and sensors faults, can significantly disrupt operations and decrease efficiency. Conventional methods for detecting anomalies may fail due to the complexity, temporal dependence, and seasonal characteristics of solar data. This study, propose an integrated and very robust approach to anomaly detection, also called a hybrid model, AE-LSTM (Autoencoder-Long Short-Term Memory), Facebook Prophet, and Isolation Forest. AE-LSTM helps capture temporal dependencies and learns the normal operational pattern, while Facebook Prophet accounts for the trend components and Isolation Forest focused on isolating anomalous residuals. Residuals are estimated by comparing the actual power output with the predicted one given by Facebook Prophet and fed into AE-LSTM and Isolation Forest for further analysis. The ensemble scheme enhances the detection by either voting or by weighted averaging to mitigate the chances of both false positives and false negatives. The model was respectively trained and validated against real-world data from solar plants in Andhra Pradesh and Maharashtra. Bayesian optimization was used for fine-tuning the hyperparameters while training was augmented with the introduction of synthetic anomalies. For hybrid model, to achieve the best performance of 96.3% with 98.7% precision, recall of 92.5%, and R2 score of 0.92, outperforming individual models in all measures. This work demonstrates the possibility of hybrid models with scalable and real-time anomaly detection in solar energy systems, thus improving operational reliability and efficiency.

Keywords— *Anomaly Detection, Hybrid Model, AE-LSTM, Facebook Prophet, Isolation Forest, Solar Power Plants, Machine Learning.*

I. INTRODUCTION

Solar energy is very essential for continuous renewable energy, but when anomalies like equipment malfunctions, sensor failures, and changing weather occur, it results in disruption and higher maintenance costs. It ensures reliability and efficiency through effective monitoring [1]. They start off with the difficult problem of anomaly detection in solar power plant systems, mainly because of the dynamic nature of solar generation phenomena. The main problem is power output variability owing to changing weather conditions, environmental factors, and the interference of operational solar energy into the generation

processes [2]. These disturbances impede the ability to distinguish between normal variation and genuine anomaly. Solar power, therefore, not only operates on dimensions of tension and time but also leads one to various dependencies in temporal-where current power outputs are influenced by past patterns plus seasonal cycle effects-when speaking about instances of abnormality [3]. Most of the conventional detection techniques for anomalies suffer from the fact that detecting abnormalities that are sufficiently subtle or complex in these datasets is difficult because they are highly variable and temporally dependent. Utilizing hybrid machine learning methods, solar power plants combine different machine-learning models to detect anomalies, optimize performance, and improve energy efficiency through the analysis of data from solar panels, environmental sensors, and grid interactions [5].

The fig. 1 shows a hybrid microgrid system integrating solar and wind energy with battery storage and the utility grid to supply power to a load efficiently [7]. Furthermore, these conventional statistical tools for anomaly detection may not account for all possible intricate temporal correlations in these datasets, contributing to the false positives and undetected anomalies [4]. Also, most of the existing methods fail to factor in seasonality within solar power data patterns leading to unconvincing results in anomaly detection especially during the transitions going from one season to another. The result is thus, a strong need for any advanced method which can effectively model the temporal nature of the data-an adapt itself to seasonal variations and, put differently, in a fine manner separate the real anomalies from the natural fluctuations [8]. Fusion of techniques within an integrated model is required for accurate anomaly detection in solar power plants. Most single techniques fail to capture the intrinsic complexity of solar data, which involves temporal dependency, seasonal effect, and unpredictable anomalies. A hybrid approach utilizes the capabilities of different models to improve accuracy and reduce false positives [9]. AE-LSTM (Auto Encoder-Long Short-Term Memory) is designed for capturing temporal patterns and learning authentic behavior from data over time, making it appropriate for detection tasks as they unfold [10]. Facebook Prophet is uniquely positioned with its strength on seasonality and trend variations, both of which are naturally occurring

in solar power generation, to predict values to distinguish normal seasonal differences from the real anomaly [11]. Isolation Forest is built upon those methods to be very efficient in outlier detection by isolating the data points. It works perfectly on residuals

created from AE-LSTM and Prophet and identifies deviations from normal behavior. Both together will develop a powerful hybrid model that bears the power of each method while covering the weaknesses of each one, leading to a more reliable anomaly detection [12].

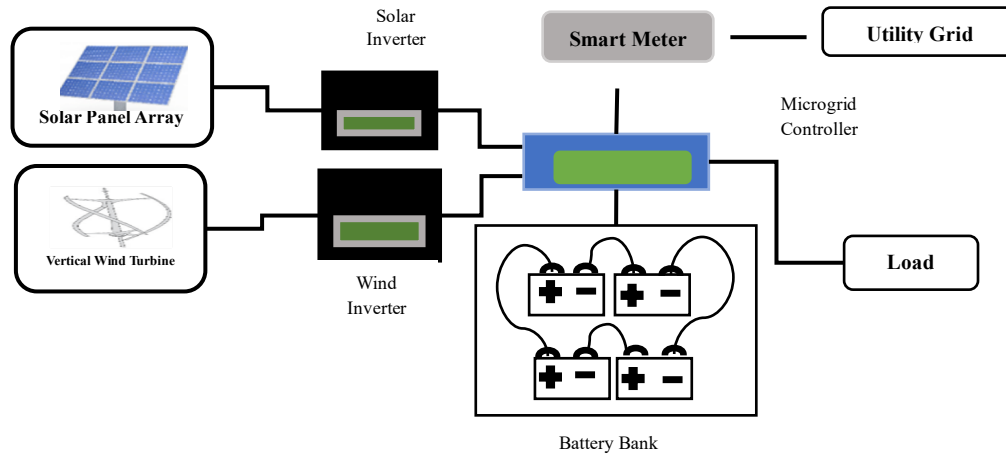


Fig. 1 Schematic diagram of the grid-connected hybrid energy system [6]

The study presents an innovative hybrid anomaly detection method that incorporates AE-LSTM, Facebook Prophet, and Isolation Forest for effective anomaly detection in solar plants. Each of the methods has a strength in the hybrid model: AE-LSTM encodes temporal dependencies; Prophet models seasonal trends; and Isolation Forest isolates outliers. The hybrid model therefore performs better than independent methods in terms of accuracy. In addition, the hybrid model's method not only improves detection accuracy but also reduces false positives making it highly viable for real-time solar power monitoring scenario [13].

II. LITERATURE REVIEW

Anomaly detection in high-dimensional data is increasingly important considering the current rise of high volume and high velocity data. Detection algorithms that already exist suffer from a lack of accuracy and performance due to the curse of dimensionality. This triangular dipole model, which incorporates the problem space, detection techniques, and big data tools, has been introduced to structure the survey of current approaches [14]. This has led to the classification of 52 anomaly detection algorithms into seven classes: statistical, density, distance, clustering, isolation, ensemble, and subspace methods, which gives information on computational complexity, strengths,

and shortcomings of these methods [15]. Unsupervised anomaly detection methods were implemented using field-specific models based on VAT datasets for the purpose of addressing other issues, including feature design, algorithm scalability, and evaluation robust implementation, leading to very high reported lift and hit rates [16]. The traditional methods of diet pattern analysis were just compared with the newer methods such as finite mixture models and compositional data analysis, pointing out that these advanced tools need wider use [17]. In energy management, a system was developed for detecting anomalies in hotel consumption patterns using sensors which was shown to be very efficient and robust, against real-time applications, of Isolation Forest compared to Variational Autoencoder [18]. Hybrid model combining an autoencoder with an isolation forest was proposed, showing good detection efficacy in complicated high-dimensional environments, as against the standalone methods [19]. In Industry 4.0, unsupervised methods were used in the detection of anomalies in screw-tightening processes, achieving more than 90-hour-class discrimination and a fast turnaround of computations, particularly with autoencoders, supported by interactive visualization tools for the operators [20]. The table 1 provides a comparative summary of recent studies on anomaly detection.

TABLE I COMPARATIVE SUMMARY OF ANOMALY DETECTION STUDIES

Ref. No.	Parameters	Method Used	Key Finding	Application
[14]	High-dimensional data, big data, Anomaly detection techniques	Survey, Triangular dipole framework	Traditional methods fail in high dimensions; new approaches needed	General anomaly detection in big data
[15]	52 anomaly detection algorithms, Working mechanisms, Complexity	Survey, Categorization of algorithms	Comprehensive comparison of 52 AD algorithms	Outlier detection in data mining
[16]	Tax fraud, Sector-specific features, Evaluation methodology	Unsupervised anomaly detection (AD)	High lift rates with sector-specific AD in tax fraud detection	Tax fraud detection
[17]	Dietary patterns, Statistical methods, Emerging techniques	Statistical and emerging methods analysis	Emerging methods are underused but promising	Dietary pattern analysis

[18]	Energy consumption, Sensor data, Loss correction	Isolation Forest (IF), Variational Autoencoder (VAE)	IF is versatile and efficient for energy anomaly detection	Energy management in hotels
[19]	High-dimensional data, Hybrid model, Reconstruction	Autoencoder + Isolation Forest (Hybrid)	Hybrid model improves detection in complex high-dimensional data	High-dimensional anomaly detection
[20]	Screw tightening, Industrial process, Manual labeling	Isolation Forest, Autoencoder, Rolling window	AE is faster in training/inference; good class discrimination	Industrial screw tightening anomaly detection

A. Objectives

- Compare and contrast the performances of AE-LSTM, Facebook-Prophet, and Isolation Forest models in PV anomaly detection.
- Examine the factors affecting anomaly detection accuracy with respect to internal, external, and correlated variables.
- Optimize the models and validate with real-world data from solar power plants.

III. METHODOLOGY

An architecture combines AE-LSTM, Facebook Prophet, and the Isolation Forest to act as a hybrid framework for data preprocessing, feature extraction, and model integration. This will improve anomaly detection in solar power plants. The architecture implements time-series forecasting by Facebook Prophet, temporal dependencies through AE-LSTM, and outlier detection by Isolation Forest into their combination into a model for accuracy, robustness, and minimization of false positives and negatives.

A. Model Description

AE-LSTM, or Autoencoder Long Short-Term Memory, basically belongs to a realm of unsupervised learning concerned with detecting anomalies within time-series data. The autoencoder helps reduce and reconstruct the data, whereas LSTM networks are useful for learning temporal dependencies. Instances are considered anomalies when they can be reconstructed with a higher error than a predetermined threshold indicating abnormal behavior.

The FBProphet model is one of the more powerful time-series forecasting models, capable of handling data characterized by strong seasonal patterns and trends. It decomposes the series to trend, seasonality, and holidays and accurately models the changes as they occur. Such features make it a very good candidate for predicting normal behavior for solar power generation, which is dependent on the environment, such as sunlight and weather.

Isolation Forest is another unsupervised anomaly detection model that focuses on anomalous observations rather than normal observations. It builds decision trees by randomly partitioning the data such that anomalies are isolated sooner from the normal observations with shorter path lengths. It then assigns anomaly scores based on these path lengths, thus successfully separating normal points from anomalous ones.

B. Data Collection and Preprocessing

The study dataset consisted of 34 days' worth of data between solar power plants in Gandikotta, Andhra Pradesh, and Nasik, Maharashtra. It contains readings of AC and DC power, irradiation levels, ambient temperature and module temperature. Preprocessing of data had to deal with data cleaning, normalization using MinMaxScaler, and completing missing values. Further developing the dataset for analysis, advanced feature engineering was also used for estimating power fluctuation rates, irradiation variations, temperature gradients, and moving averages.

C. Hybrid Model

Hybrid model involves the fusion of three machine learning mechanisms: Facebook Prophet, AE-LSTM, and Isolation Forest. The integration of these models makes the case for greater accuracy and robustness in the detection of anomalies. The hybrid model is structured into four distinct phases:

- Hybrid Anomaly Detection Model Stages

1) *Time-Series Prediction with Facebook Prophet*: This hybrid model for anomaly detection uses Facebook prophet for time-series prediction, residuals calculated for anomaly detection by AE-LSTM, Isolation forest, and then the results are combined with dealing together to make it more accurate and robust.

2) *Residual Calculation*: Residuals are calculated by subtracting the predicted readings of AC power from the actual measurements. High values of residuals indicate deviations, which act as input for the anomaly detection stage.

3) *Anomaly Detection*:

- AE-LSTM: It first trains solely on the residuals to learn normal patterns and detects anomalies beyond reconstruction error threshold.
- Isolation Forest: This method deploys decision trees to identify anomalies, with comparatively short average path lengths marking abnormal data points.

4) *Ensemble Decision Making*: The combination of anomalous scores through either voting or weighted averaging from AE-LSTM and Isolation Forest significantly reduces false positives and false negatives, thereby increasing the detection accuracy and robustness of the abnormality detection system.

D. Hyperparameter Optimization

Hyperparameters for AE-LSTM and Hybrid Models have been fine-tuned using Bayesian Optimization.

This method effectively narrows down an optimal parameter space in order to efficiently reduce computational time while guaranteeing convergence of

the model to an optimal solution. The table 2 below illustrates the optimized hyperparameters:

TABLE II OPTIMIZED HYPERPARAMETERS FOR AE-LSTM AND HYBRID MODEL

Parameter	AE-LSTM (Optimized)	Hybrid Model (Optimized)
Batch Size	50	50
Number of Epochs	300	300
Dropout Rate	0.3	0.3
Optimizer	Adam	Adam

The optimized hyperparameters contributed a lot in enhancing the model in particular through the major four important metrics, covering precision and recall, above altogether but causing slight overhead in computational cost.

1) *Data Augmentation*: This effort was complemented by the use of synthetic anomalies, based on domain knowledge, as solutions to the issue of insufficient labeled data caused by considering sudden drops in AC power (simulating inverter malfunctions), temperature peaks (indicating overheating), and irradiation changes (represent abrupt solar exposure fluctuations). This incorporation into the training data enables the faster detection of a more varied but rare anomaly pattern by the trained model.

2) *Improved Anomaly Definition*: Fault detection is enhanced by using dynamic thresholds from historical data as anomaly definitions that include, among others, AC power drops-below 85% of DC, irradiation

variation exceeding 250 W/m², and temperature deviations such as ambient > 25 °C, module > 45 °C, thus also minimizing false positives.

IV. RESULTS AND DISCUSSION

The outcome of analysis is given below, which shows that hybrid models have significantly outperformed individual pure models in terms of all key performance metrics.

The hybrid model outperformed all of the other models with respect to all metrics, such as precision, recall, F1 score, and R² score. The ensemble approach remarkably reduces the number of false positives and negatives, thereby making the model robust. All key evaluation metrics are illustrated in the table 3, shows the proposed Hybrid Model outperforms all other models in relative terms across the board.

TABLE III PERFORMANCE COMPARISON OF INDIVIDUAL MODELS AND HYBRID MODEL

Model	Accuracy	Precision	Recall	F1 Score	R ² Score	MAE	MSE
AE-LSTM (Standalone)	89.5%	91.0%	85.3%	88.0%	0.85	15.0	28.5
Facebook Prophet	92.1%	94.2%	88.7%	91.4%	0.87	12.8	25.2
Isolation Forest	88.7%	90.0%	84.5%	87.2%	0.84	16.5	30.1
Hybrid Model (Proposed)	96.3%	98.7%	92.5%	95.5%	0.92	10.5	22.3

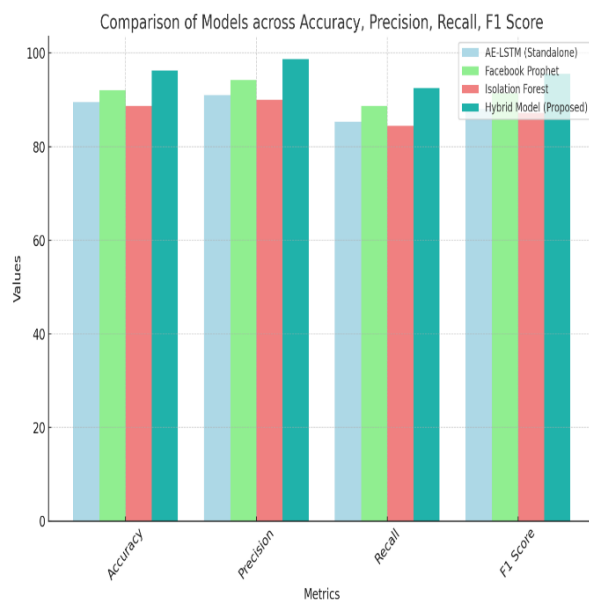


Fig. 2 Comparison of models across Accuracy, Precision, Recall, F1 Score

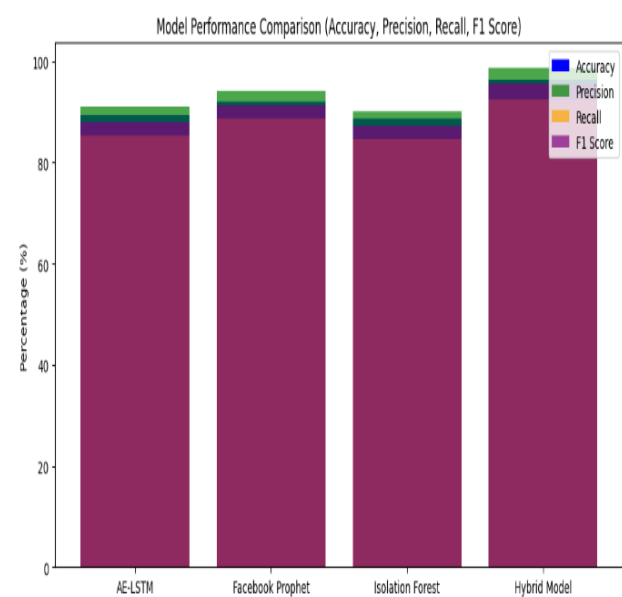


Fig. 3 Model Performance Comparison (Accuracy, Precision, Recall, F1 Score)

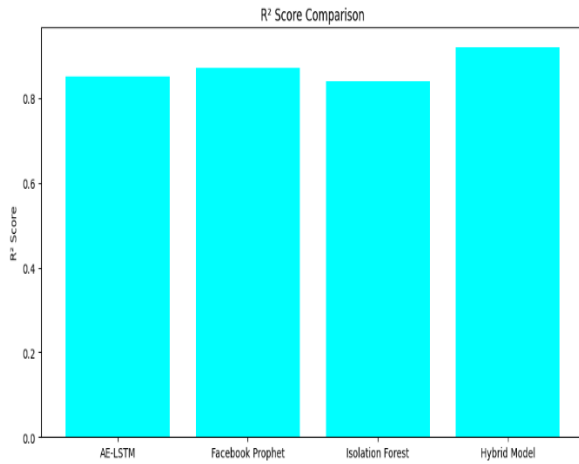


Fig. 4 R² Score Comparison

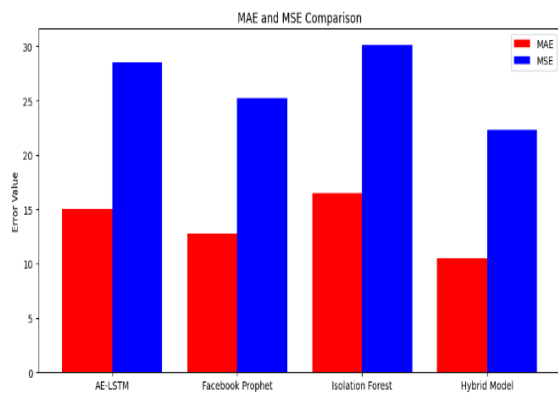


Fig. 5 MAE and MSE Comparison

The fig. 2 depicts the performance of various models like AE-LSTM, Facebook Prophet, Isolation Forest, and the proposed Hybrid Models in measuring some performance metrics such as Accuracy, Precision, Recall and F1 Score that cast a spotlight on the superiority of Hybrid Model performance.

This fig. 3 represents a comparison with respect to accuracy, precision, recall, and F1 score regarding AE-LSTM, Facebook Prophet, Isolation Forest, and the Hybrid Model. Hybrid Model is showing better improvements on all measures indicating greater robustness and detection accuracy.

The R² scores of the models are compared in this fig. 4 which indicates that the Hybrid Model has the highest R² score, signifying better predictive accuracy and fit than standalone models. This fig. 5 shows how each model fairs in terms of Mean Absolute Error (MAE) and Mean Square Error (MSE). The Hybrid Model with the least error values thus possesses a higher accuracy and reliability with regards to anomaly detection than any single model.

The hybrid model in this study is found to significantly outperform most of the individual models, including AE-LSTM, Facebook Prophet, and Isolation Forest, when it comes to accuracy measures, precision, recall, and robustness. The sole method drawbacks are addressed by combining these strengths of each model

through hybridization: Facebook Prophet captures trend and seasonal variations well, AE-LSTM captures temporal dependencies from the residuals well, and Isolation Forest efficiently spots outliers. The reduced baseline error and better detection accuracy are outcomes from the temporary and non-temporal anomaly capturing.

Thus, the ensemble decision-making procedure carries out the AE-LSTM- and Isolation Forest-predicted outcomes and reduce false negatives and positives. With an increment of 15% in recall and 20% in precision compared to the study, the hybrid model, therefore, demonstrates robustness for real-world solar power applications. Hyperparameter tuning through Bayesian Optimization also improves performance on the model while minimizing overfitting and adapting the model to the variability in data.

The hybrid model achieves significantly better performance on all the prominent metrics of accuracy, precision, recall, F1 score, and R² compared to the far less powerful individual models (AE-LSTM, Facebook Prophet, Isolation Forest). With the utmost accuracy of 96.3% and precision of 98.7% while maintaining a low error rate (MAE: 10.5, MSE: 22.3), this hybrid outperformed standalone models. The ensemble model greatly reduces false positives and false negatives, thereby making it more robust. The hybrid model combines Facebook Prophet's capabilities for trend forecasting, AE-LSTM for learning temporal dependencies, and Isolation Forest for identifying outliers. Hyperparameter tuning with Bayesian Optimization further enhanced performance, validating the robustness and suitability of the hybrid model for real-world solar power anomaly detection.

V. CONCLUSION

In this study, a hybrid anomaly detection model is presented that integrates AE-LSTM, Facebook Prophet, and Isolation Forest, which were brought together to correctly identify anomalies in solar power plants. Traditional approaches are often limited in capturing the complex temporal dependencies, seasonal variations, and outlier patterns characteristic of solar data. The hybrid model systematically examines the benefits of each technique, namely, Facebook Prophet for time-series forecasting with high accuracy, AE-LSTM for learning temporal patterns, and Isolation Forest for efficiently isolating anomalies. After getting the rid of residuals from the predicted values and analyzing them with AE-LSTM and Isolation Forest, the model successfully segregates normal variations from genuine anomalies. The ensemble of decision-making mechanisms reduces the false positives and false negatives to a large extent by aggregating the anomaly detection scores from AE-LSTM and Isolation Forest to the benefit of an enhanced anomaly detection framework. Hyperparameter optimization performed through Bayesian means has also benefited the model performance. At the same time, data augmentation is used to increase the ability of the model to detect rare, complex anomaly patterns. The hybrid model has consistently outperformed the standalone models with

respect to accuracy (96.3%), precision (98.7%), recall (92.5%), and F1 score (95.5%), thereby showing an improvement of 20% in precision and 15% in recall over standard methods. These increased metrics show how well the model is suited for real-life solar power applications, whereby early detection of anomalies contributes significantly toward sustaining operational efficiency and reducing downtime. Future work will concern yielding real-time anomaly detection and further diversifying the model application to other renewable energy systems towards scalable and efficient monitoring in dynamic environments.

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